

Siddhartha Gudipudi

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Education

Iowa State University

Bachelor of Science in Computer Science

Expected: Dec 2025

GPA: 3.70

Work Experience

Software Automation Co-op, Corteva Agriscience

Jan 2025 – August 2025

- Developed and modernized projects by migrating from **VB.NET** to **C#(.NET9)** and upgrading **Python** projects with efficient libraries to enhance performance and maintainability of code base, replacing old and outdated libraries and practices.
- Developed and modified multiple enterprise applications using **.NET/C#**, python for efficient seed management and reducing seed wastage by ~28%. Used **Nunit** for testing the applications and **SQLite** for data storage and management.
- Designed and developed a **Raspberry Pi 5** based automated seed counting system, using **.NET9/C#**, python and trained **YOLO** deep learning model. Utilized **Linux** to run the scripts in pi 5 and streamed the video using **OpenCV** and **GStreamer**.
- Validated **REST APIs** with **SwaggerUI** and executed **CRUD** operations to create, update, verify seed factors, Id's and records.
- Led collaboration with cross-functional teams to gather requirements and deliver tailored technical solutions as required.

Software Development Intern, RDI – Thinix

May 2024 – Dec 2024

- Developed and maintained features in the IStatus application using **Angular/TypeScript, C#**, and frameworks like **Entity** framework and **ASP.NET, Microsoft SQL Server Management Studio (SSMS)** for data management.
- Built and optimized security report feature, leveraging **SQL** queries to manage and manipulate data, and used **Azure DevOps** for **CI/CD pipelines, version control**, documentation, automating testing, and streamlining the deployment process.
- Thoroughly tested the application with **MSTest** and **Jest**, improving test coverage by 20% for reliable and bug free code base.
- Conducted code reviews and participated in **sprint meetings** to enforce best practice, improve code quality and maintainability

Student IT Client Support, DoR - Iowa State University

March 2023 – present

- Provided technical support to resolve software/hardware issues and collaborated via **ServiceNow, Confluence**, and **JIRA** with IT teams and departments to ensure minimal downtime and an optimal campus-wide user experience.
- Developed **python scripts** for cleaning and splitting the PC's data and used **Linux scripting** for day to day administrative tasks.

Student Programmer, Iowa State University

Dec 2023 - May 2024

- Developed and debugged features for Stone Chenn Lab (virtual chemistry lab), used **TypeScript, React, Java**, and **Python**
- Built interactive experiment simulations, dynamic graphs using **matplotlib**, and automated chemical calculations using **NumPy**
- Deployed the application with **Docker**, ensuring scalability and streamlined operations and managed code with **Git**

Projects

AI Backend recommendation system – Go, SQLite, Docker, LLM's, Raspberry Pi 5

- Designed and implemented an end-to-end AI recommendation engine for IdeaNova's InPlay Media Portal, powering personalized media and shopping suggestions by combining user watch history, shopping activity, and behavioral data.
- Built Go (chi), REST APIs, SQLite (indexed), in-process caching, and Dockerized deployment for fast, reliable recommendations.
- Researched and benchmarked open-source LLMs with Ollama to select a compact, low-compute model to run on rasp pi 5.
- Worked with a 4-member team and client in weekly reviews, adapting to evolving requirements and delivering milestones.

Bird Watch app - Spring Boot, JAVA, Android Studio, Web Sockets, MySQL, Google Maps API, TensorFlow, RESTful API, GIT

- Designed a mobile app utilizing neural network-powered machine learning (TensorFlow) to identify birds from a database of 55,000 images. Users can capture bird photos, receive detailed species info, and track locations.
- Utilized Java, Spring Boot, TensorFlow, and MySQL to develop backend services, CRUDL operations, and REST APIs, integrated Google maps API for real me location tracking of the birds, sharing real insights with friends' group and species identification.
- Automated deployment and updates using GitLab CI/CD pipelines, achieving 80% test coverage with JUnit and Mockito.
- Used Android Studio for frontend interface development, led troubleshooting to reduce downtime by 25%, and conducted code reviews to enforce best practices in machine learning, API development, and to maintain high quality and efficient code.

HackathonF23(2nd Position) AstroStory - Next.js, TypeScript, Node.js, Docker, OpenAI Vision API, DALL-E, GPT-4

- Developed AstroStory, a creative storytelling app for cosmic adventurous stories with Next.js + TypeScript and Node.js.
- Integrated OpenAI Vision API for image analysis, DALL-E for image generation, and GPT-4 for narrative creation, offering users a variety of tones and genres. Built infrastructure with Docker for reliability and scalability ensuring consistency on platforms.

Cy-Commerce - MERN stack, CRUD

- Developed a full-stack e-commerce platform using MongoDB, Express.js, React, and Node.js with features for product catalog management, adding items to cart, checkout, order processing, and user logging feature with log creation and reactions.

Skills

Programming Languages: Python, Java, C#, C, C++, TypeScript, Golang, SQL, bash scripting.

Frameworks: HTML5, CSS, Spring Boot, .NET Core, AngularJS, Next.JS, Node.JS, React.JS, TensorFlow, Gatsby.js, Django, Entity

Tools: Git, Azure DevOps, AWS, Docker, API's

Databases: SQL, MongoDB